

AKYAR, Russian Federation

WMO: 350370

Lat: 51.873N

Lon: 58.223E

Elev: 341

StdP: 97.29

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.3	-26.4	-32.6	0.2	-29.2	-29.7	0.3	-26.2	11.1	-20.4	9.8	-15.8	2.8	300	0.630

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	32.4	18.0	30.4	17.4	28.7	16.9	19.6	28.0	18.7	27.2	17.9	26.1	3.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.0	12.6	22.2	15.9	11.8	21.5	14.9	11.0	20.9	57.3	27.8	54.4	27.2	51.7	26.0	24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.5	9.3	8.1	DB	-31.8	34.4	3.2	1.7	-34.1	35.6	-36.0	36.6	-37.8	37.6	-40.2	38.8	(4)
(5)			WB	-31.9	20.9	3.1	1.5	-34.1	22.0	-35.9	22.9	-37.6	23.8	-39.9	24.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.8	-14.4	-13.2	-6.6	5.1	13.9	18.7	20.1	19.1	12.6	4.6	-4.3	-11.2	(6)	
	DBStd	13.76	6.64	6.55	5.43	6.32	4.54	4.27	3.57	4.42	4.45	5.18	6.02	6.65	(7)	
	HDD10.0	3394	756	649	514	171	16	1	0	1	23	179	428	657	(8)	
	HDD18.3	5528	1014	882	773	396	149	47	22	46	179	427	678	915	(9)	
	CDD10.0	1132	0	0	0	25	137	262	314	283	100	11	0	0	(10)	
	CDD18.3	224	0	0	0	0	12	58	78	69	7	0	0	0	(11)	
	CDH23.3	2562	0	0	0	7	191	668	839	766	90	1	0	0	(12)	
	CDH26.7	886	0	0	0	1	33	240	289	296	27	0	0	0	(13)	
(14) Wind	WSAvg	3.6	3.4	3.8	3.9	4.1	3.8	3.6	3.6	3.3	3.2	3.5	3.8	3.6	(14)	
(15) Precipitation	PrecAvg	364	27	19	29	29	32	42	48	38	22	25	25	28	(15)	
	PrecMax	624	54	61	63	78	63	111	124	150	54	47	43	60	(16)	
	PrecMin	227	6	2	1	3	4	19	10	5	1	1	1	8	(17)	
	PrecStd	103	12	13	18	20	17	23	31	34	16	13	13	14	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.3	1.2	4.9	24.6	29.2	34.1	34.9	34.5	30.2	21.9	9.2	1.4	(19)	
		MCWB	-0.3	0.1	2.0	12.7	15.1	18.1	18.0	18.0	15.8	11.9	6.1	0.7	(20)	
	2%	DB	-1.3	-0.4	2.8	21.5	27.5	31.5	31.8	32.1	25.6	18.0	6.1	0.2	(21)	
		MCWB	-2.0	-1.1	1.0	11.5	14.6	17.9	18.4	17.8	13.8	10.3	4.4	-0.3	(22)	
	5%	DB	-2.6	-1.7	1.6	18.2	25.6	29.6	29.7	30.0	23.1	15.2	4.2	-0.8	(23)	
		MCWB	-3.2	-2.3	0.4	10.0	14.3	17.4	17.6	17.4	13.3	8.3	2.7	-1.5	(24)	
	10%	DB	-4.4	-3.6	0.7	15.0	23.2	27.4	27.9	27.8	20.4	12.6	2.7	-2.3	(25)	
		MCWB	-5.0	-4.3	-0.3	8.6	13.4	16.7	17.3	16.6	12.2	7.5	1.6	-3.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.2	0.3	2.7	14.1	16.9	20.5	20.9	20.5	17.0	12.6	7.1	0.8	(27)	
		MCDB	0.1	0.8	3.9	22.8	25.3	29.1	28.9	28.9	27.9	19.6	7.8	1.3	(28)	
	2%	WB	-1.8	-1.1	1.5	12.1	16.0	19.1	19.8	19.1	15.3	11.0	4.5	-0.3	(29)	
		MCDB	-1.3	-0.5	2.5	19.4	24.6	28.4	27.7	28.5	22.8	16.4	5.9	0.0	(30)	
	5%	WB	-3.2	-2.4	0.6	10.7	15.1	18.2	18.8	18.1	14.0	9.3	2.9	-1.4	(31)	
		MCDB	-2.6	-1.7	1.5	17.1	23.5	27.2	26.4	27.3	20.6	13.8	4.1	-0.9	(32)	
	10%	WB	-4.9	-4.3	-0.2	9.1	14.1	17.4	18.0	17.3	12.7	7.9	1.6	-3.0	(33)	
		MCDB	-4.3	-3.6	0.7	14.5	22.1	25.5	25.3	25.6	19.3	11.5	2.7	-2.3	(34)	
(35) Mean Daily Temperature Range	MDBR		6.8	7.6	7.7	9.7	12.2	12.1	11.5	12.0	11.1	8.8	5.6	6.4	(35)	
	5% DB	MCDBR	6.0	6.5	6.9	14.6	15.4	15.0	14.4	14.9	15.1	14.5	6.8	5.9	(36)	
		MCWBR	5.6	6.0	5.8	6.8	6.0	5.4	4.8	4.9	6.4	7.7	5.4	5.8	(37)	
	5% WB	MCDBR	5.7	6.7	6.1	13.3	13.6	13.3	12.0	13.1	12.9	12.6	6.2	5.8	(38)	
		MCWBR	5.6	6.2	5.4	6.8	6.3	5.1	4.8	4.8	6.2	7.1	5.2	5.7	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.292	0.320	0.375	0.418	0.403	0.398	0.409	0.429	0.376	0.350	0.322	0.289	(40)	
	taud		2.080	2.023	1.947	2.077	2.238	2.317	2.291	2.206	2.313	2.289	2.182	2.076	(41)	
	Ebn at Noon		677	767	792	812	849	857	838	796	801	739	639	603	(42)	
	Edh at Noon		78	110	146	146	130	122	123	127	102	85	71	66	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.23	2.36	3.78	4.68	5.77	6.38	6.03	5.09	3.59	1.95	0.90	0.79	(44)	
	RadStd		0.12	0.15	0.39	0.46	0.50	0.47	0.36	0.39	0.36	0.33	0.12	0.14	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+105	+84

Nomenclature: See separate page